HSCovMamba: CNN–Mamba Gated-Fusion Network for Hyperspectral Image Classification

Chunxiao Li, Xueli Geng, Jia Shi, Ming Zhang, Liwei Song, and Yuanyuan Zhang

***Abstract*—Hyperspectral images (HSIs) comprise hundreds of spectral channels, enabling fine-grained material identification and pixel-level scene understanding. The HSI classification task assigns each pixel to a semantic category by exploiting spectral signatures and spatial context. However, convolutional neural network (CNN)-based approaches are limited by local receptive fields, while Transformer-based approaches suffer from high computational cost. Although Mamba-based approaches provide efficient long-range contextual modeling, they still insufficiently address the key challenge of preserving local structural fidelity under fine-grained HSI classification. To address these limitations, we propose HSCovMamba, a structure-preserving CNN–Mamba framework for efficient HSI classification. Specifically, deformable local focused convolution (DLF-Conv) performs lightweight local feature extraction through deformation-aware and channel-adaptive modeling, improving sensitivity to irregular textures and fine-grained local variations. Meanwhile, the bi-directional dual-gated fusion decoder (BDGF-Decoder) progressively aligns high-level semantics with low-level spatial details through bidirectional cross-scale fusion, leading to more boundary-consistent HSI representations. Experiments on benchmark datasets demonstrate that HSCovMamba delivers superior classification accuracy and efficiency compared with current state-of-the-art (SOTA) methods, validating its effectiveness for lightweight HSI analysis.**

***Index Terms*—Bi-directional dual-gated fusion (BDGF), deformable local focused convolution (DLF-Conv), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL images (HSIs) comprise hundreds of contiguous spectral bands, offering rich spectral and spatial information for precise identification of ground materials and objects [1], [2], [3]. HSI classification assigns a class label to each pixel, enhancing the analysis and interpretation of remote sensing data [4]. As a core remote sensing task, it is widely used in agricultural management, urban planning, and mineral exploration [5], [6]. Traditional HSI classification approaches, such as support vector machines [7] and random forests [8], [9], primarily rely on shallow spectral features. This limits their ability to exploit comprehensive spectral–spatial information, reducing classification accuracy and adaptability across diverse scenarios.

With the progress of deep learning, numerous convolutional neural networks (CNNs) have been widely investigated for HSI classification [10], [11], [12], [13], [14], [15], [16], [17], [18], [19]. By employing hierarchical convolutional structures, CNN-based approaches are capable of automatically extracting discriminative spatial features, achieving significantly higher performance compared with traditional methods. Nonetheless, the intrinsic limitation of local receptive fields hinders CNNs from effectively modeling long-range dependencies within hyperspectral data. Although 3-D convolutional networks (3-D CNNs) [13], [14], [15], [16] enable joint modeling of spatial and spectral dimensions to a certain extent, the capability to capture global correlations remains insufficient.

To overcome these limitations, Transformer-based approaches [20], [21], [22], [23], [24] have been developed, leveraging self-attention to capture global dependencies and achieving impressive performance in HSI classification tasks. However, the quadratic growth of computational complexity with input size makes self-attention prohibitively expensive for high-dimensional HSI data, constraining its use in large-scale or real-time applications.

Recently, Mamba has provided an alternative paradigm through its core state space model (SSM), which effectively alleviates the computational bottlenecks of Transformers. SSMs model contextual information via state transitions rather than computing all pairwise token relationships, resulting in substantially improved efficiency. The Mamba architecture [25], as the latest advancement in SSMs, incorporates a selective state space mechanism that dynamically retains or discards information according to contextual relevance, thereby enabling more efficient long-term dependency modeling. Despite its strength in capturing global information, Mamba is less effective at modeling fine-grained local details. CNNs, on the other hand, naturally excel in local feature extraction with superior computational efficiency.

Mamba-based HSI classification methods typically follow two design paradigms, i.e., Transformer-inspired

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Chunxiao Li is with the School of Computer Science and Technology, Xidian University, Xi'an, Shaanxi 710071, China, and also with the 27th Research Institute of China Electronics Technology Group Corporation, Zhengzhou 450047, China (e-mail: xiaoxiao870114@126.com).

Xueli Geng and Jia Shi are with the School of Artificial Intelligence, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: gengxueli@xidian.edu.cn; ljxswj@163.com).

Ming Zhang is with the 27th Research Institute of China Electronics Technology Group Corporation, Zhengzhou 450047, China.

Liwei Song is with Chaoyue Technology Co., Ltd., Jinan, Shandong 250104, China

Yuanyuan Zhang is with the School of Physics, Xidian University, Xi'an, Shaanxi 710071, China.

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spectral–spatial integration strategies [26], [27], [28] and Mamba's linear multidirectional scanning mechanism for sequence modeling [29]. Although these methods are effective in capturing long-range contextual dependencies, they either weaken the computational efficiency advantage of Mamba or fail to adequately preserve fine-grained local structures. For HSI classification, however, accurate recognition depends not only on global contextual reasoning but also on preserving fine-grained local structures, such as subtle texture variations, irregular geometric patterns, and clear class boundaries. Existing CNN–Mamba hybrids [30] partially alleviate this issue by combining convolutional locality with Mamba-based global modeling, yet they still lack an explicit mechanism for structure-preserving local geometric modeling and cross-scale semantic-detail reconciliation. As a result, local texture shifts and boundary transitions are often insufficiently modeled, leading to blurred predictions in fine-grained regions.

To address this limitation, we propose HSCovMamba, a CNN–Mamba gated-fusion network for efficient and structure-preserving HSI classification. HSCovMamba follows a dual-stream hybrid UNet architecture. In the encoder, local-global feature extraction blocks couple deformable local focused convolution (DLF-Conv) with Mamba to jointly model local geometric fidelity and long-range contextual dependencies. In the decoder, a bi-directional dual-gated fusion (BDGF) mechanism progressively reconciles high-level semantics with low-level spatial details across scales, leading to more discriminative and boundary-consistent hyperspectral representations. The contributions of this article are outlined as follows.

1) We propose HSCovMamba, a structure-preserving CNN–Mamba framework for HSI classification, which effectively balances global contextual modeling, local structural fidelity, and computational efficiency.
2) We introduce a DLF-Conv module for lightweight local structural modeling. Through adaptive local feature modulation, DLF-Conv efficiently captures irregular textures, geometric variations, and fine-scale spatial patterns in HSI with low computational cost.
3) We develop a bi-directional dual-gated fusion decoder (BDGF-Decoder) for cross-scale semantic-detail reconciliation. By progressively aligning high-level semantic cues with low-level structural details across scales, it improves boundary consistency and fine-grained discrimination in HSI classification.

The remainder of this article is organized as follows. Section II presents a concise overview of existing studies. Section III describes the proposed HSCovMamba. Section IV provides an empirical assessment of the proposed method, supported by ablation studies and extensive comparisons with current competitive methods. Finally, Section V concludes this article with key insights and elaborates on its application scenarios.

II. Related Work

The study of HSI classification has gained increasing research interest, driven by its wide application potential and the difficulties posed by high-dimensional spectral information coupled with intricate spatial structures. One key challenge in HSI analysis is the high spectral redundancy, which motivates band selection to reduce input dimensionality and improve efficiency. For example, MR-Selection uses meta-reinforcement learning to enable zero-shot band selection on unseen HSIs by transferring knowledge across datasets [3], while M^3BS [31] presents a novel multiteacher multiobjective meta-learning framework to learn transferable band-selection knowledge. Additionally, HSI models often face spectral-induced distribution discrepancies, which S4DL [32] effectively addresses through shift-sensitive spatial–spectral disentangling that separates invariant and specific components. In these methods, deep learning techniques have played a pivotal role in advancing this field, enabling models to automatically derive abstract, discriminative features and capture global contextual information [33]. Existing deep models are typically organized into three representative families according to their architectures: CNN-based, Transformer-based, and Mamba-based approaches.

A. CNN-Based Approaches

To fully exploit the rich information contained in HSI, mainstream CNN-based approaches primarily include 2-D CNNs and 3-D CNNs [10]. A 2-D CNN extracts spatial features from HSI patches by treating each spectral band as an independent channel, similar to the RGB channels in conventional images [11], [12]. Although spatial contextual information is effectively represented, this approach fails to fully exploit the comprehensive spectral information embedded within HSIs. In contrast, 3-D CNNs extend convolutions to the spectral dimension, enabling joint processing of spatial and spectral information. By treating spectral bands as an additional dimension, it simultaneously learns spatial relationships among pixels and spectral dependencies across bands. For example, Zhong et al. [13] proposed an end-to-end spatial–spectral 3-D CNN with residual connections to enable deeper network architectures, while Zhang et al. [14] introduced a 3-D Dense CNN that leverages dense connections for efficient cross-layer information flow in HSI classification. These architectures offer more comprehensive feature representations and substantially improve classification performance, but they also introduce challenges such as high computational cost and a tendency toward overfitting [15], [16]. To further enhance HSI classification performance, several architectural variants have been proposed. To capture spatial information at multiple granularities, multiscale CNNs extract features from diverse resolution levels [17]. Integrating attention mechanisms further enables CNNs to concentrate on critical and discriminative features, boosting classification robustness in complex environments [18], [19].

CNN-based approaches have shown promising performance over traditional methods. However, their dependence on local receptive fields and large amounts of labeled data limits their ability to model long-term complex spatial dependencies between pixels, which hinders the capture of global contextual information. Consequently, more advanced network designs are required to effectively capture global contextual information.

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B. Transformer-Based Approaches

To overcome the limitations of CNNs that rely on local receptive fields, researchers have progressively adapted Transformer architectures to jointly model spectral and spatial dimensions in HSI data. For example, Hong et al. [20] proposed SpectralFormer, which uses spectral tokenization and band-specific attention to effectively capture interband dependencies while preserving spatial coherence. The cross-attention spatial–spectral Transformer (CASST) [21] uses a dual-path architecture to separately model spatial and spectral sequences and incorporates a spatial–spectral cross-attention module with weighted sharing to fuse features within Transformer blocks. Researchers have explored combining CNNs and Transformers to effectively leverage the complementary strengths of both models in feature extraction. Zhang et al. [22] propose CTMixer, a hybrid framework that combines CNNs and Vision Transformers to capture local–global features in HSI. Zhao et al. [23] present GSC-ViT, a lightweight transformer for HSI classification that combines group-wise separable convolution and multihead self-attention to efficiently capture local spectral–spatial and global features, and employ a simplified pointwise layer to enhance feature fusion. In addition, Geng et al. [24] developed a knowledge-aware geometric contourlet semantic learning framework that combines the multiscale and multidirectional representations of the contourlet transform with Transformer-based feature learning, enhancing robustness and interpretability. These advances collectively demonstrate the versatility and effectiveness of Transformer-based architectures in addressing the challenges of HSI classification.

Despite this progress, Transformer-based approaches for HSI classification still face notable challenges. The self-attention mechanism incurs high computational and memory costs, which are especially problematic for HSI due to the large number of spectral channels that amplify the processing burden. These complexities make it difficult to deploy Transformer-based models in large-scale or resource-constrained scenarios. As a result, increasing attention has been directed toward designing more efficient network architectures, lightweight attention mechanisms, and hybrid designs that can maintain the benefits of global context modeling while improving scalability and reducing the computational demands inherent to HSI data.

C. Mamba-Based Approaches

Recently, the structured SSM (S4) has made significant advancements in sequence data analysis, enabling optimized linear computational costs when processing long sequences. Mamba builds on S4 by introducing a selection mechanism that dynamically focuses on input-dependent features, enhancing both efficiency and performance. Furthermore, Mamba integrates hardware-aware design to increase computational efficiency, making it more effective than traditional transformer models in handling complex tasks.

Researchers have explored applying the Mamba architecture to the field of HSI classification, leveraging Mamba's advantage in capturing global contextual information to enhance the efficiency of classification tasks [29]. Spectralmamba [26] comprises a spectral–spatial token generation module and stacked Mamba blocks, where spatial and spectral tokens are processed separately and then fused through a feature enhancement module, enabling effective information integration and improved classification performance. He et al. [27] converted the HSI cube into 3-D spectral–spatial tokens and introduced a selective scanning mechanism to efficiently capture global spectral–spatial relationships. Sheng et al. [30] introduced DualMamba for HSI classification, an architecture that integrates the strengths of CNN and Mamba to improve computational efficiency while capturing intricate spectral–spatial relationships. However, it has limitations, particularly in multidirectional scanning, where exploiting spectral information fully is difficult, and redundancy occurs. Ahmad et al. [28] enhanced HSI classification by integrating wavelet transformation with the spatial–spectral Mamba.

Nevertheless, existing Mamba-based HSI classification methods mainly focus on spectral–spatial dependency modeling while paying relatively limited attention to structure-preserving local modeling and cross-scale semantic-detail coordination. As a result, fine-grained textures, irregular geometric variations, and boundary transitions are still insufficiently modeled in complex HSI scenes. In contrast, HSCovMamba is designed to preserve local structural fidelity while maintaining efficient long-range contextual modeling, providing a structure-preserving local–global collaborative framework for HSI classification.

III. Proposed Method

This section details the implementation of HSCovMamba. First, the preliminary knowledge related to Mamba is discussed. Then, an overall overview of HSCovMamba is presented, followed by a detailed analysis of its two core components: DLF-Conv and BDGF-Decoder.

A. Preliminaries: Mamba

The Mamba model is constructed to capture long-range dependencies with linear complexity, making it highly suitable for large-scale data processing. It overcomes the high computational cost that Transformer models face when processing long-range dependencies. The core module of Mamba is SS2D (2D-selective-scan), which efficiently models input data by introducing linear variations in the latent states. The system dynamics for the input $x(t)$ and output $y(t)$ are given by

$$\mathbf{h}'(t) = A\mathbf{h}(t) + Bx(t)$$
$$y(t) = C\mathbf{h}(t) + Dx(t)$$

where A and B are key matrices for modeling long-range dependencies, with A governing the evolution of the latent states and B modeling the input response. The matrix C defines the mapping from the latent states to the output, while D adjusts the direct influence of the input on the output. These linear relations allow the SS2D module to efficiently capture long-range dependencies, ensuring low computational overhead.

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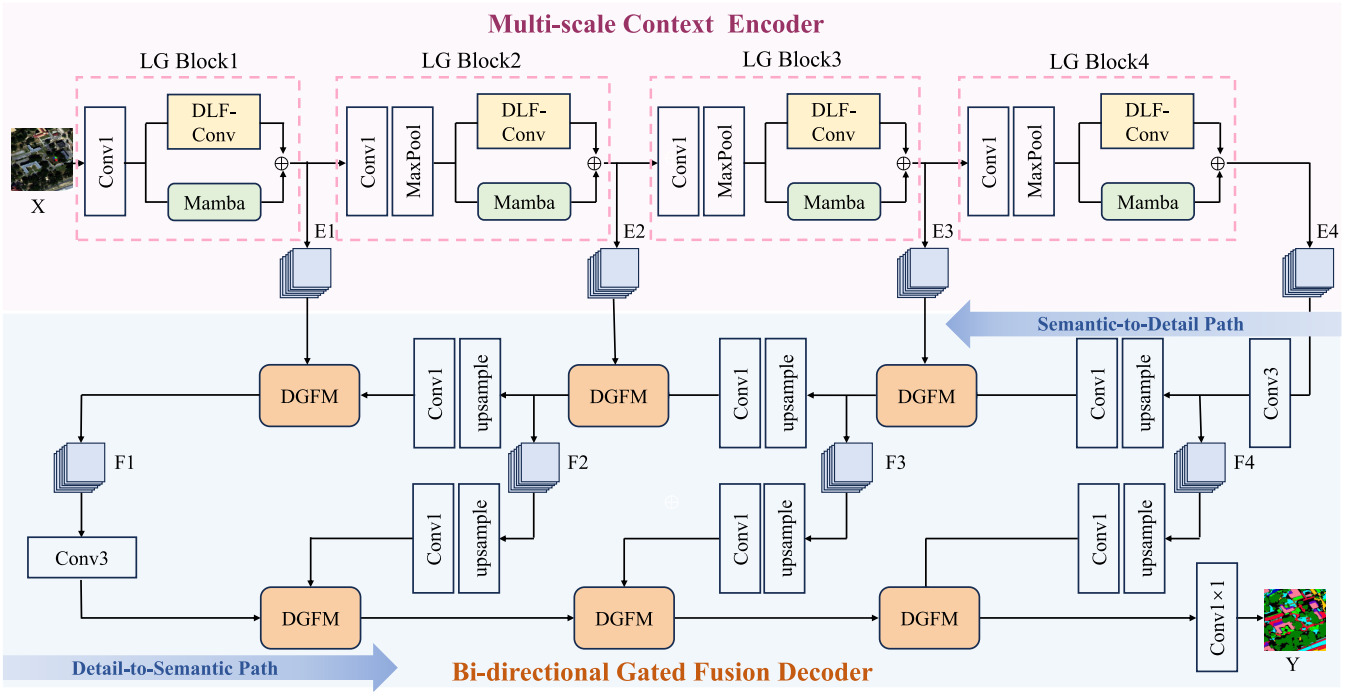


Fig. 1. Architecture of the proposed HSCovMamba.

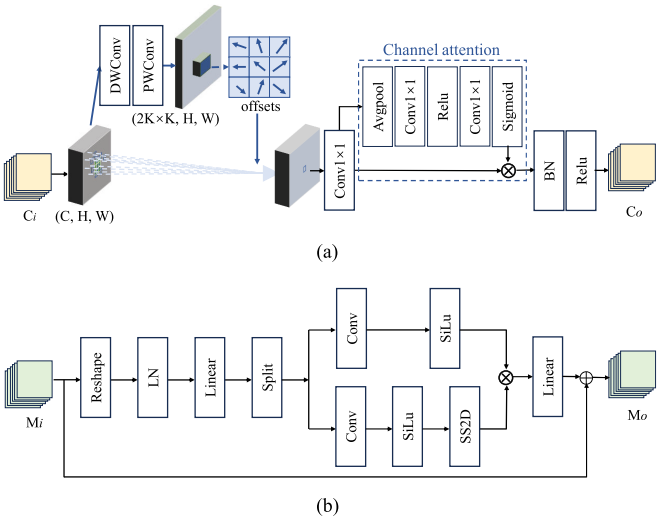


Fig. 2. Detailed illustration of feature extraction architecture. (a) DLF-Conv: local feature extraction for capturing fine-grained spatial patterns. (b) Mamba: global feature extraction for aggregating long-range dependencies and contextual information.

B. Overview of HSCovMamba

As illustrated in Fig. 1, HSCovMamba comprises a multiscale context encoder and a BDGF-Decoder. Given an HSI $\mathbf{X} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$, where C_{in} represents the channel dimension and H and W denote the spatial resolution, the multiscale context encoder processes the image through four local–global feature extraction blocks (LG Blocks) to generate hierarchical features. The first LG Block differs from the subsequent three. Specifically, in the first LG Block, Conv1 is followed by parallel DLF-Conv and Mamba operations, where the diagram for DLF-Conv is shown in Fig. 2(a) and the diagram for Mamba is shown in Fig. 2(b). In the subsequent blocks, an additional Maxpool operation is applied after Conv1. Here, Conv1 denotes a 1×1 convolution, followed by batch normalization (BN) and ReLU. DLF-Conv captures local geometric deformations and texture details with low computational cost, and the Mamba layer models long-range dependencies and global contextual information with linear complexity. Maxpool operation serves to downsample the feature map. As a result, the outputs of these blocks are represented as $\{E1, E2, E3, E4\}$, and their sizes are 1, 1/2, 1/4, and 1/8 of the size of the HSI. The hierarchical features are then fed into the BDGF-Decoder for multiscale feature fusion, producing the final pixel-wise classification map $\mathbf{Y}$

$$\mathbf{Y} = \text{BDGF-Decoder}(E_1, E_2, E_3, E_4).$$

The network is optimized with a cross-entropy loss, which promotes accurate pixel-wise classification by increasing the predicted probability of the correct class for each pixel. Given the predicted pixel classification map (Y) and the ground truth map (T), the cross-entropy loss is defined as

$$\mathcal{L}_{\text{CE}} = -\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} t_{hwc} \log(y_{hwc})$$

where H and W are the spatial dimensions of the image; C is the number of classes; t_{hwc} is the ground truth label for the hth row, wth column, and cth class of the pixel in the image (with $t_{hwc} \in 0, 1$); and y_{hwc} is the predicted probability for the same pixel and class (with $y_{hwc} \in [0, 1]$).

C. Deformable Local Focused Convolution

HSIs pose two coupled challenges for local representation learning: severe spectral redundancy and complex spatial

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structures with irregular local geometry and subtle boundary deformation. Conventional convolutions with fixed receptive fields are limited in adapting to such geometric variations, often failing to preserve fine-grained textures and boundary cues in HSI scenes. Although standard deformable convolutions provide adaptive sampling capability, directly applying dense deformable kernels to high-dimensional HSI features introduces considerable computational overhead. To address this issue, we introduce DLF-Conv, a lightweight geometry-aware local modeling module that captures irregular local deformations and discriminative texture patterns at low cost, thereby producing more focused local representations for HSI classification. The detailed procedures of DLF-Conv are shown in Fig. 2(a).

Given an input feature map $\mathbf{C}_i \in \mathbb{R}^{C_{\text{in}} \times H \times W}$, the DLF-Conv module first applies a depthwise convolution followed by a pointwise convolution to generate spatial offsets for the deformable convolution. The offsets are computed as

$$\Delta \mathbf{p} = \text{PWConv}(\text{DWConv}(\mathbf{C_i})) \in \mathbb{R}^{2K^2 \times H \times W}$$

where K denotes the kernel size and $\Delta \mathbf{p} = \{\Delta p_k\}_{k=1}^{K^2}$ with each $\Delta p_k \in \mathbb{R}^2$ representing a learnable 2-D displacement for the kth sample location.

The deformable convolution is performed independently on each input channel. For an output at spatial location p_0 and channel c, the channel-wise deformable response is defined as

$$U_c(p_0) = \sum_{k=1}^{K^2} w_{c,k}\, \mathbf{X}_c(p_0 + p_k + \Delta p_{c,k}(p_0))$$

where p_0 denotes the kernel center, $w_{c,k}$ is the depthwise convolution weight for channel c, p_k denotes the relative position of the kth kernel element with respect to p_0, and $\Delta p_{c,k}(p_0)$ is the learned offset for channel c and kernel element k at location p_0.

Subsequently, a pointwise convolution is applied to perform channel mixing

$$\mathbf{V} = \text{Conv}_{\text{pw}}(\mathbf{U})$$

where Conv_{pw} denotes the 1×1 convolution. To further enhance discriminative capability, a channel attention mechanism is employed to reweight the output channels. The channel attention vector is computed as

$$\mathbf{a} = \sigma\left(\text{Conv}_2\left(\delta\left(\text{Conv}_1\left(\text{GAP}\left(\mathbf{V}\right)\right)\right)\right)\right)$$

where $\text{GAP}(\cdot)$ denotes global average pooling, $\delta(\cdot)$ is the ReLU activation function, $\sigma(\cdot)$ is the sigmoid function, and Conv_1 and Conv_2 are 1×1 convolutional layers used for channel interaction. The final block output is

$$\mathbf{C_o} = \text{ReLU}(\text{BN}(\mathbf{V} \otimes \mathbf{a}))$$

where $\otimes$ denotes element-wise channel-wise multiplication and $\text{BN}(\cdot)$ represents batch normalization.

This design yields a module that preserves the high adaptability of deformable convolutions while retaining the efficiency of depthwise separable operations. The integrated channel attention further amplifies informative channels, allowing the network to capture localized geometric deformations with minimal computational burden, thereby improving both texture representation and fine-grained recognition performance.

1) Lightweight Design: The DLF-Conv block attains efficiency by leveraging a depthwise separable design. The deformable operation is executed channel-wise (depthwise) rather than using a full $C_{\text{in}} \times C_{\text{out}}$ deformable kernel. Let the number of channels be C. For a conventional dense deformable convolution with kernel size K, the parameter count (ignoring biases) scales as

$$\text{Params}_{\text{dense}} = C_{\text{in}} \cdot C_{\text{out}} \cdot K^2.$$

In contrast, the DLF-Conv parameterization consists primarily of: 1) depthwise deformable weights $C_{\text{in}} \cdot K^2$; 2) pointwise mixing $C_{\text{in}} \cdot C_{\text{out}}$; and 3) a small offset predictor (a depthwise 3×3 plus a 1×1 projection) and a lightweight channel attention ($\approx 2C_{\text{out}}/r$ params). Hence, the dominant terms are

$$\text{Params}_{\text{DLF-Conv}} \approx C_{\text{in}} \cdot K^2 + C_{\text{in}} \cdot C_{\text{out}} + O(C_{\text{in}})$$

which is typically much smaller than $\text{Params}_{\text{dense}}$ when K is small and C_{in} is moderate. The depthwise deformable design also reduces multiply–accumulate operations (MACs) relative to dense deformable convolutions, making DLF-Conv attractive for resource-constrained inference.

D. Bi-Directional Dual-Gated Fusion Decoder

Existing methods primarily utilize a unidirectional (top-down) fusion of semantic information from deep to shallow layers. While this enhances shallow-layer semantics, the deep features do not receive real-time feedback or correction during the fusion process. Moreover, simple fusion operations such as addition or concatenation indiscriminately mix features and easily introduce shallow noise into semantic features. This issue is particularly harmful for HSI classification, where accurate prediction requires both semantic discrimination and precise boundary preservation. To address this, we propose a BDGF-Decoder. This decoder jointly propagates semantic and spatial information through two complementary pathways. It incorporates a dual-gated feature modulation (DGFM) module, which adaptively determines the relevance of spatial details to the global context. This mechanism enriches the semantic maps with accurate structural information while effectively suppressing background noise.

1) Semantic-to-Detail Fusion Path: High-level semantic features from the deepest encoder layer contain rich category information but suffer from coarse spatial resolution. In contrast, low-level features offer detailed structural information but often contain irrelevant semantic noise. This semantic-to-detail path is designed to leverage these high-level priors as a conceptual anchor to guide the integration of fine-grained details. Specifically, it progressively upsamples high-level semantic features and selectively merges them with lower-level encoder representations, enriching fine-grained spatial details with relevant contextual semantics. Formally, given the multilevel encoder features (E_1, E_2, E_3, E_4)

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from shallow to deep layers, the fusion process is defined as

$$\begin{aligned} F_4 &= \text{Conv3}\left(E_4\right) \\ F_3 &= \text{DGFM}\left(\text{Conv1}\left(\text{Up}\left(F_4\right)\right), E_3\right) \\ F_2 &= \text{DGFM}\left(\text{Conv1}\left(\text{Up}\left(F_3\right)\right), E_2\right) \\ F_1 &= \text{DGFM}\left(\text{Conv1}\left(\text{Up}\left(F_2\right)\right), E_1\right) \end{aligned}$$

where Conv3(·) represents a 3×3 convolution followed by BN and ReLU, Up(·) denotes an upsampling operation implemented by bilinear interpolation, and Conv1(·) represents a 1×1 convolution followed by BN and ReLU. DGFM adaptively selects and reweights complementary information from the two input features. By progressively propagating contextual semantics to shallower layers, the decoder can effectively disambiguate low-level spatial noise and ensure that fine-grained representations are anchored to the correct semantic regions.

2) Detail-to-Semantic Fusion Path: This pathway allows the decoder to refine higher level semantics using structural priors from the lower layers, resulting in spatially coherent and contextually consistent representations. It serves to validate the semantic features by using the fine-grained details to adjust and correct any potential inaccuracies in the semantic understanding. Starting from the detail-enhanced feature F_1, this reverse fusion process can be formulated as

$$\begin{aligned} G_1 &= \text{Conv3}\left(F_1\right) \\ G_2 &= \text{DGFM}\left(G_1, \text{Conv1}\left(\text{Up}\left(F_2\right)\right)\right. \\ G_3 &= \text{DGFM}\left(G_2, \text{Conv1}\left(\text{Up}\left(F_3\right)\right)\right. \\ G_4 &= \text{DGFM}\left(G_3, \text{Conv1}\left(\text{Up}\left(F_4\right)\right)\right. . \end{aligned}$$

This pathway enables the decoder to adaptively refine higher level semantics by incorporating structural priors from lower layers, resulting in spatially coherent and contextually consistent representations.

The final output feature is produced by applying a lightweight prediction head $\psi(\cdot)$ to the fused feature G_4

$$\mathbf{Y} = \psi(G_4)$$

where $\psi(\cdot)$ consists of a 1×1 convolutional projection, with the output channels corresponding to the number of target classes.

3) DGFM Module: Formally, given a high-level feature map $\mathbf{H_1} \in \mathbb{R}^{C \times H \times W}$ and a low-level feature map $\mathbf{H_2} \in \mathbb{R}^{C \times H \times W}$, the gating weights are computed as

$$[\mathbf{W}_1, \mathbf{W}_2] = \sigma(\text{Conv}_2(\delta(\text{Conv}_1(\text{Concat}(\mathbf{H_1}, \mathbf{H_2})))))$$

where $\delta(\cdot)$ and $\sigma(\cdot)$ denote the ReLU and sigmoid functions, respectively. Two weights $\mathbf{W}_1$ and $\mathbf{W}_2$ modulate the contribution of $\mathbf{H_1}$ and $\mathbf{H_2}$, respectively. The gated features are fused by

$$\mathbf{H} = \text{Conv}_3(\mathbf{W}_1 \otimes \mathbf{H_1} + \mathbf{W}_2 \otimes \mathbf{H_2})$$

where $\text{Conv}_i(\cdot)$ denotes a 1×1 convolution. The detailed structure of the DGFM module is shown in Fig. 3. This dual-gated mechanism suppresses irrelevant information while emphasizing complementary and discriminative ones, thereby facilitating precise feature alignment across scales.

IV. Experiments

A. Experimental Settings

1) Datasets: Extensive experiments on three benchmark HSI datasets—PaviaU, Salinas, and HongHu—are conducted to evaluate the effectiveness and robustness of the proposed method. The number of training and testing samples for the three datasets is listed in Table I.

a) Paviau dataset: It was collected above the Pavia University area in Italy. It has a spatial resolution of 1.3 m/pixel and spatial dimensions of 610×340 and includes 103 spectral bands covering wavelengths from 430 to 860 nm, which consists of 9 categories.

b) Salinas dataset: It was collected by the AVIRIS sensor over Salinas Valley, California, with a spatial resolution of 3.7 m/pixel. It contains 204 spectral bands covering wavelengths from 400 to 2500 nm, with spatial dimensions of 512×217. The dataset is classified into 16 categories.

c) Honghu dataset: It was collected using the Headwall Nano-Hyperspec sensor over the agricultural-park boundary in Honghu City, Hubei Province, with a spatial resolution of 0.043 m/pixel. The dataset includes 270 spectral bands spanning wavelengths from 400 to 1000 nm and has a spatial dimension of 940×475. It is divided into 22 classes.

2) Implementation Details: HSCovMamba is developed with the PyTorch framework, and both training and testing are performed on an HP-Z840 workstation. This setup includes a GeForce GTX TITAN X GPU with 12 GB of memory and runs on a 64-bit Ubuntu 18.04.5 operating system. During training, the network parameters are initialized using the Xavier method [34]. The Adam optimizer [35] is employed with an initial learning rate of 1×10^{-2} and a cosine decay schedule. To balance predictive accuracy with computational efficiency, the number of epochs is fixed at 50 for each dataset. For each benchmark, the training set X_t is formed by randomly selecting 20 labeled samples per class, with the remaining labeled samples assigned to the test set. To alleviate overfitting, uniform data augmentation is applied across all evaluated methods, consisting of random flips and rotations of the training samples. All methods are evaluated under the same data split, preprocessing pipeline, training/testing protocol, and implementation environment. For method-specific settings, such as patch size and batch size, the standard configurations reported in the original papers are followed to ensure faithful implementation.

3) Evaluation Metrics: We employ quantitative metrics and qualitative analysis to assess the classification effectiveness of various methods. For qualitative evaluation, we visually evaluate classification performance by comparing the classification maps produced by various methods against the ground truth map, facilitating qualitative evaluation of the results. For quantitative evaluation, overall accuracy (OA), average accuracy (AA), and Kappa are employed as the primary evaluation metrics, which are widely recognized in HSI classification for jointly measuring overall correctness, class-balanced performance, and agreement beyond chance.

Let a C-class classification result be summarized by a confusion matrix $\mathbf{N} = (N_{ij}) \in \mathbb{N}^{C \times C}$, where N_{ij} denotes the

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TABLE I
LAND COVER CLASSES AND SAMPLE COUNTS (TRAIN/TEST) ACROSS PAVIAU, SALINAS, AND HONGHU DATASETS

ID	PaviaU		Salinas		HongHu	
	Class	Train/Test	Class	Train/Test	Class	Train/Test
1	Asphalt	20/6611	Brocoli_green_weeds_1	20/1989	Red roof	20/14021
2	Meadows	20/18629	Brocoli_green_weeds_2	20/3706	Road	20/3492
3	Gravel	20/2079	Fallow	20/1956	Bare soil	20/21801
4	Trees	20/3044	Fallow_rough_plow	20/1374	Cotton	20/163265
5	Metal sheets	20/1325	Fallow_smooth	20/2658	Cotton firewood	20/6198
6	Bare Soil	20/5009	Stubble	20/3939	Rape	20/44537
7	Bitumen	20/1310	Celery	20/3559	Chinese cabbage	20/24083
8	Bricks	20/3662	Grapes_untrained	20/11251	Pakchoi	20/4034
9	Shadows	20/927	Soil_vinyard_develop	20/6183	Cabbage	20/10799
10	–	–	Corn_senesced_green_weeds	20/3258	Tuber mustard	20/12374
11	–	–	Lettuce_romaine_4wk	20/1048	Brassica parachinensis	20/10995
12	–	–	Lettuce_romaine_5wk	20/1907	Brassica chinensis	20/8934
13	–	–	Lettuce_romaine_6wk	20/896	Small Brassica chinensis	20/22487
14	–	–	Lettuce_romaine_7wk	20/1050	Lactuca sativa	20/7336
15	–	–	Vinyard_untrained	20/7248	Celtuce	20/982
16	–	–	Vinyard_vertical_trellis	20/1787	Film covered lettuce	20/7242
17	–	–	–	–	Romaine lettuce	20/2990
18	–	–	–	–	Carrot	20/3197
19	–	–	–	–	White radish	20/8692
20	–	–	–	–	Garlic sprout	20/3466
21	–	–	–	–	Broad bean	20/1308
22	–	–	–	–	Tree	20/4020

number of test samples with ground-truth class i predicted as class j. Define the total number of test samples

$$N = \sum_{i=1}^{C} \sum_{j=1}^{C} N_{ij}$$

and the row/column marginals

$$N_{i\cdot} = \sum_{j=1}^{C} N_{ij} \quad \text{(support of class } i\text{)}$$

$$N_{\cdot j} = \sum_{i=1}^{C} N_{ij} \quad \text{(predicted count of class } j\text{)}.$$

a) Overall accuracy: OA measures the proportion of correctly classified samples over the entire test set

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} N_{ii}}{N}.$$

While widely reported, OA is sensitive to class imbalance—dominant classes in HSI datasets can disproportionately influence this metric.

b) Average accuracy: AA is the macro-averaged per-class accuracy. For class i, the per-class (recall) accuracy is

$$A_i = \frac{N_{ii}}{N_{i\cdot}} \quad \text{for } N_{i\cdot} > 0$$

and AA is computed as

$$\mathrm{AA} = \frac{1}{C} \sum_{i=1}^{C} A_i = \frac{1}{C} \sum_{i=1}^{C} \frac{N_{ii}}{N_{i\cdot}}.$$

AA treats all classes equally and is thus more informative under long-tailed label distributions common in HSI. When a class has zero support ($N_{i\cdot} = 0$), it should be explicitly reported and either excluded from the macro-average or handled via a predefined convention.

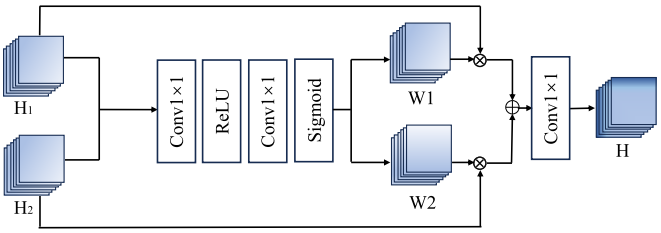


Fig. 3. Detailed illustration of DGFM module.

c) Kappa: Kappa quantifies agreement between predictions and ground truth beyond chance. Let the observed agreement be

$$p_o = \mathrm{OA} = \frac{\sum_{i=1}^{C} N_{ii}}{N}$$

and the expected agreement by chance, under independent marginals, be

$$p_e = \sum_{i=1}^{C} \left(\frac{N_{i\cdot}}{N}\right)\left(\frac{N_{\cdot i}}{N}\right) = \frac{1}{N^2} \sum_{i=1}^{C} N_{i\cdot} N_{\cdot i}.$$

Then,

$$\kappa = \frac{p_o - p_e}{1 - p_e}.$$

B. Performance Comparison

To evaluate the performance of the proposed HSCov-Mamba, we compare it against nine state-of-the-art (SOTA)

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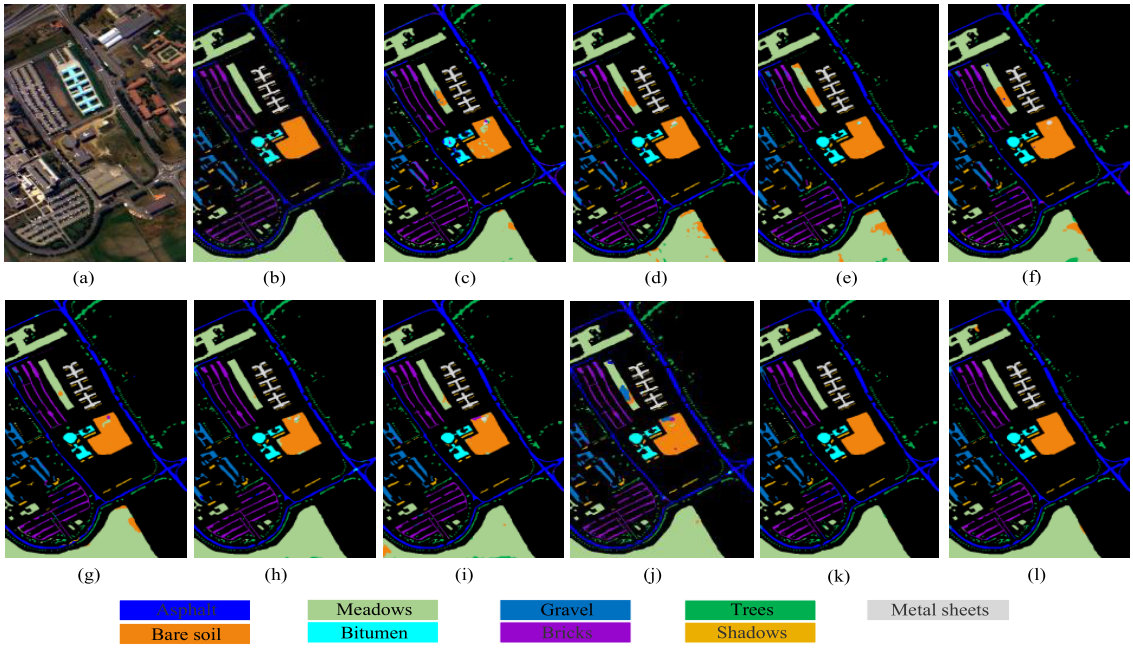


Fig. 4. Classification maps of different methods on the PaviaU dataset. (a) Hyperspectral pseudo-color image. (b) Ground truth. (c) SSAN. (d) A2S2KResnet. (e) CVSSN. (f) SSTN. (g) MASSFormer. (h) KGCSL. (i) SS-Mamba. (j) S2Mamba. (k) MambaHSI. (l) HSCovMamba.

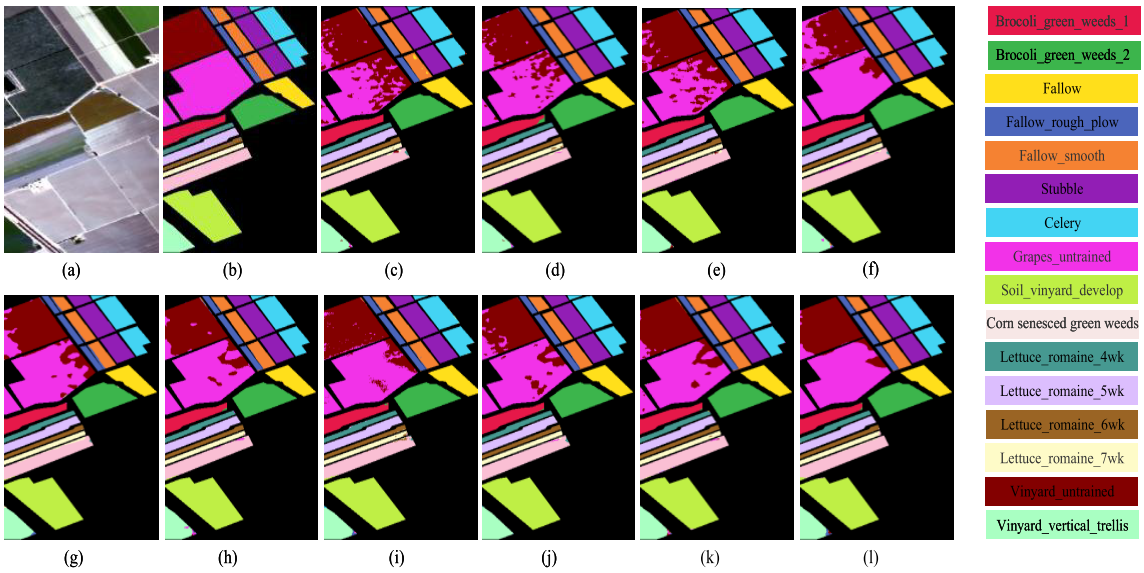


Fig. 5. Classification maps of different methods on the Salinas dataset. (a) Hyperspectral pseudo-color image. (b) Ground truth. (c) SSAN. (d) A2S2KResnet. (e) CVSSN. (f) SSTN. (g) MASSFormer. (h) KGCSL. (i) SS-Mamba. (j) S2Mamba. (k) MambaHSI. (l) HSCovMamba.

methods grouped into three categories: 1) CNN-based methods: SSAN [16], A2S2KResNet [36], and CVSSN [37]; 2) Transformer-based methods: SSTN [38], MASSFormer [39], and KGCSL [24]; and 3) Mamba-based methods: SS-Mamba [40], S2Mamba [41], and MambaHSI [42].

Tables II–IV present the quantitative results of different competing methods on the three hyperspectral datasets, with the best performance in each row emphasized in boldface. The classification maps obtained by all compared methods on the three datasets are illustrated in Figs. 4–6.

1) Results on PaviaU Dataset: Table II summarizes the results of different methods on the PaviaU dataset. HSCovMamba demonstrates superior performance over other methods, reaching the highest OA, AA, and Kappa scores, with scores of 98.50%, 98.67%, and 98.01%, respectively. Compared to the second-best performing method, MambaHSI (OA = 97.27% and Kappa = 96.55%), HSCovMamba achieves an OA improvement of 1.23%, an AA improvement of 1.42%, and a Kappa improvement of 1.46%. Based on class-level evaluation, HSCovMamba achieves superior accuracy in nearly

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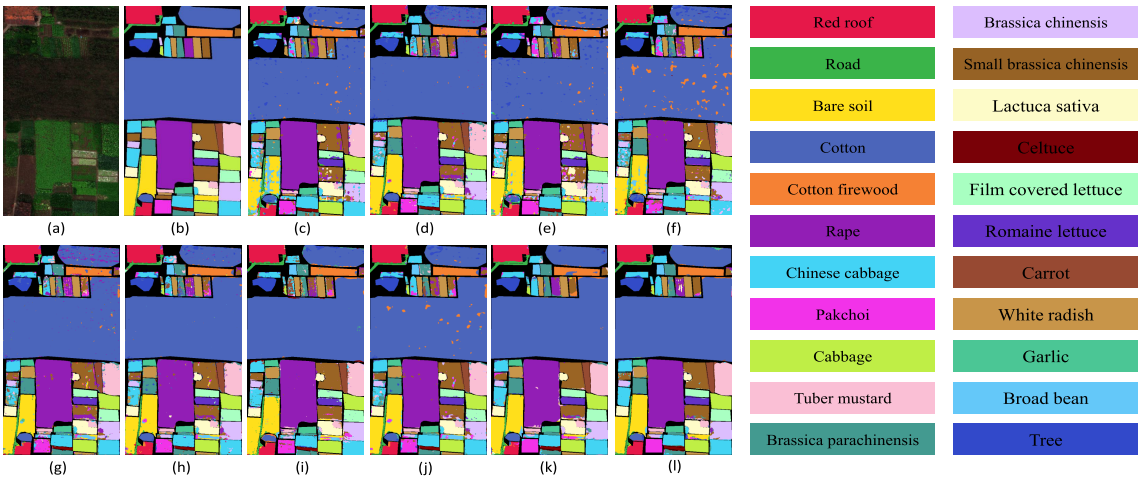


Fig. 6. Classification maps of different methods on the HongHu dataset. (a) Hyperspectral pseudo-color image. (b) Ground truth. (c) SSAN. (d) A2S2KResnet. (e) CVSSN. (f) SSTN. (g) MASSFormer. (h) KGCSL. (i) SS-Mamba. (j) S2Mamba. (k) MambaHSI. (l) HSCovMamba.

TABLE II
QUANTITATIVE CLASSIFICATION ACCURACY (%) ON THE PAVIAU DATASET

ID	CNN-based			Transformer-based			Mamba-based			
	SSAN	A2S2KResnet	CVSSN	SSTN	MASSFormer	KGCSL	SS-Mamba	S2Mamba	MambaHSI	HSCovMamba
1	86.12	88.45	91.12	93.05	96.23	97.81	**98.42**	97.75	95.11	96.70
2	94.98	95.32	93.91	90.25	95.44	96.55	96.71	95.93	**99.35**	99.16
3	87.83	93.15	93.78	89.04	94.62	97.12	98.79	96.06	99.85	**100.0**
4	92.84	**97.66**	96.03	92.04	92.17	96.41	96.74	95.53	90.88	97.10
5	99.75	**100.0**	99.73	99.53	99.91	**100.0**	**100.0**	**100.0**	99.77	**100.0**
6	90.54	91.96	90.02	97.41	96.05	94.93	93.85	94.72	**100.0**	99.98
7	94.07	98.74	98.83	98.95	99.87	**100.0**	**100.0**	99.37	**100.0**	**100.0**
8	84.22	88.45	81.16	94.11	90.83	95.02	94.85	96.34	90.53	**95.24**
9	99.85	99.74	99.04	98.65	98.24	**99.92**	99.67	99.11	99.78	99.89
OA	91.94	93.37	92.58	92.81	95.17	96.85	97.09	96.38	97.27	**98.50**
AA	92.24	94.83	93.73	94.78	95.93	97.53	97.67	97.20	97.25	**98.67**
Kappa	89.33	91.34	90.64	90.79	93.88	95.88	96.03	95.23	96.55	**98.01**

half of all categories, specifically in four out of nine classes, while maintaining high accuracy (greater than 96%) in the remaining categories.

Fig. 4 presents the classification maps produced by different methods and hyperspectral pseudo-color image, where the map generated by the proposed HSCovMamba exhibits the closest alignment with the ground truth. In contrast, as shown in Fig. 4(c)–(j), other methods exhibit considerable misclassifications at class boundaries and exhibit salt-and-pepper noise, resulting in discontinuous regions within the class areas. Furthermore, the Bricks class, which occupies a small and discontinuous area, is incorrectly classified by MambaHSI [Fig. 4(k)] but accurately recognized by the proposed model. This superior performance benefits from the optimized architecture—where the DLF-Conv module captures fine-grained textures and spatial variations, and the BDGF-Decoder effectively balances semantic integrity and local detail, mitigating boundary blurring and fragmentation.

2) Results on Salinas Dataset: Table III presents the quantitative performance of all methods on the Salinas dataset. The proposed HSCovMamba achieves superior performance over competing methods, showing improvements of 2.15%, 0.20%, and 1.27% in OA, AA, and Kappa, respectively, compared to the second-best method, MambaHSI. Additionally, in the prediction of the Grapes untrained class, HSCovMamba significantly outperforms other methods, with an improvement of 2.93% over the second-best method. This highlights its advantage in fine-grained classification, particularly in effectively differentiating structurally similar land cover types. This gain stems from the DLF-Conv module, which employs adaptive sampling to emphasize detailed local representations, thereby minimizing class-level confusion induced by the degradation of geometric features.

Fig. 5 shows the hyperspectral pseudo-color image and compares the ground truth with the classification maps generated by different methods on the Salinas dataset. HSCovMamba clearly outperforms other methods in accurately classifying the Vinyard_untrained and Grapes_untrained classes. Additionally, HSCovMamba demonstrates superior performance in capturing edge details, particularly for the

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TABLE III
QUANTITATIVE CLASSIFICATION ACCURACY (%) ON THE SALINAS DATASET

ID	CNN-based			Transformer-based			Mamba-based			
	SSAN	A2S2KResnet	CVSSN	SSTN	MASSFormer	KGCSL	SS-Mamba	S2Mamba	MambaHSI	HSCovMamba
1	**99.99**	99.17	99.22	99.46	99.91	99.68	99.34	99.44	96.88	99.14
2	**100.0**	**100.0**	99.95	99.94	99.90	**100.0**	**100.0**	100.0	99.78	**100.0**
3	99.73	99.99	**100.0**	**100.0**	99.99	**100.0**	99.69	**100.0**	98.05	**100.0**
4	95.02	96.32	99.57	98.48	94.55	98.50	99.70	**99.78**	99.19	99.63
5	96.51	99.35	**99.58**	96.05	95.41	99.02	97.02	97.13	96.76	98.79
6	99.99	**100.0**	99.97	98.23	99.91	**100.0**	**100.0**	**100.0**	99.64	**100.0**
7	99.61	99.89	99.47	99.69	**100.0**	99.93	98.87	99.80	99.46	99.88
8	69.70	79.80	76.24	83.19	84.12	91.31	88.62	88.63	82.84	**94.24**
9	**100.0**	99.98	99.91	**100.0**	**100.0**	99.97	99.93	**100.0**	99.41	99.82
10	94.57	96.84	96.11	95.28	96.90	**98.09**	95.48	97.41	97.02	96.50
11	99.99	99.86	99.22	99.08	**100.0**	98.15	98.56	**100.0**	99.23	99.80
12	99.97	**100.0**	99.97	99.71	99.73	**100.0**	99.94	99.94	**100.0**	99.42
13	**100.0**	99.98	99.96	99.70	99.78	**100.0**	98.77	99.88	**100.0**	99.55
14	98.75	97.75	99.67	99.73	99.89	98.40	97.61	**100.0**	99.04	98.00
15	75.97	92.03	71.32	62.13	88.42	85.97	**96.09**	86.89	94.10	94.82
16	98.49	98.84	98.75	97.77	99.09	97.69	96.81	**99.71**	97.31	95.97
OA	93.52	95.49	94.18	93.53	95.65	96.02	96.34	95.50	96.40	**98.55**
AA	89.69	94.26	90.76	90.58	94.50	95.93	97.90	98.04	98.27	**98.47**
Kappa	89.28	93.63	90.36	89.84	93.70	95.08	95.63	94.99	96.00	**97.27**

TABLE IV
LAND COVER CLASSIFICATION ACCURACY ON THE HONGHU DATASET (%)

ID	CNN-based			Transformer-based			Mamba-based			
	SSAN	A2S2KResnet	CVSSN	SSTN	MASSFormer	KGCSL	SS-Mamba	S2Mamba	MambaHSI	HSCovMamba
1	88.34	94.61	93.96	92.15	94.92	94.73	**95.40**	94.92	89.02	94.37
2	93.93	95.50	92.15	**97.94**	87.43	91.45	96.18	95.98	97.14	94.98
3	58.55	88.98	81.21	73.90	87.92	86.05	86.29	82.90	88.64	**92.10**
4	96.73	97.18	95.09	93.80	96.79	98.74	97.54	97.12	**99.37**	98.95
5	89.13	93.48	89.87	93.87	88.30	93.77	96.22	95.54	89.34	**98.95**
6	97.16	95.79	95.05	95.05	96.01	95.12	94.62	94.43	96.82	**97.40**
7	78.30	82.78	72.77	75.34	77.37	79.32	80.81	85.35	89.67	**92.24**
8	84.98	92.18	94.97	66.71	79.80	91.11	86.56	84.66	98.07	**99.73**
9	91.10	**99.16**	93.80	97.98	98.19	98.43	96.89	98.19	98.66	98.75
10	85.19	88.42	86.04	90.00	88.63	91.01	89.63	92.57	92.84	**95.02**
11	82.95	85.77	84.95	87.69	84.09	77.79	84.66	84.05	**90.49**	88.64
12	72.86	79.61	83.25	73.53	77.15	77.52	83.06	75.43	81.91	**84.50**
13	76.54	72.32	72.58	79.77	68.80	75.91	81.19	77.17	78.19	**84.28**
14	88.79	95.29	91.21	92.43	94.67	97.31	95.96	97.06	**97.91**	97.59
15	97.45	94.78	96.54	97.66	96.33	95.80	96.60	97.85	97.76	**100.0**
16	89.80	97.78	92.89	94.50	98.66	98.59	96.20	96.87	**98.74**	97.86
17	91.91	97.86	95.38	95.85	98.46	97.69	90.81	99.19	**99.77**	98.66
18	97.03	96.71	99.59	96.56	91.37	97.37	98.49	98.40	**100.0**	**100.0**
19	95.43	89.40	**97.87**	91.03	86.64	90.23	90.51	94.43	93.59	93.66
20	96.13	94.83	97.95	94.66	93.36	94.83	**98.09**	97.65	94.11	97.66
21	96.56	99.46	93.65	98.62	97.09	99.77	99.77	**100.0**	**100.0**	**100.0**
22	96.07	95.82	96.99	98.23	90.75	98.78	99.98	99.15	**100.0**	**100.0**
OA	89.90	92.80	90.59	89.94	91.50	93.07	93.15	92.87	94.97	**96.01**
AA	88.41	92.17	90.81	89.88	89.67	91.88	92.52	92.68	94.18	**95.70**
Kappa	87.30	90.93	88.23	87.47	89.27	91.23	91.40	91.06	93.64	**94.96**

Corn_senesced_green_weeds and Vinyard_vertical_trellis classes, outperforming other methods in boundary accuracy. This is attributed to the DLF-Conv and BDGF-Decoder, which enable fine-grained local feature extraction and multiscale feature fusion.

3) Results on HongHu Dataset: Table IV presents the detailed quantitative results on the HongHu dataset. The proposed HSCovMamba surpasses all competing methods, achieving gains of 1.04%, 1.52%, and 1.32% in OA, AA, and Kappa, respectively, compared with the second-best method,

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MambaHSI. In terms of per-class classification accuracy, HSCovMamba achieves the highest accuracy on most classes while maintaining consistently high accuracy on the remaining ones. Notably, in the Bare soil and Small brassica chinensis classes, the model achieves significantly superior results, exceeding the second-best method by more than 3%.

Fig. 6 presents the hyperspectral pseudo-color image and shows the predicted classification maps of different methods alongside the ground-truth map on the HongHu dataset. Visually, the map generated by the proposed HSCovMamba aligns more closely with the ground truth compared to other methods. Pronounced salt-and-pepper noise is observed in Fig. 6(c)–(g) and (j). Moreover, Fig. 6(h), (i), and (k) exhibits substantial discrepancies relative to ours in the interior regions and boundary texture details of the Broad bean and Cotton firewood classes. The superiority of the proposed method in these aspects stems from the DLF-Conv module's precise capture of fine-grained local features and the BDGF decoder's guidance on global semantic consistency.

C. Ablation and Analysis

This section provides an ablation analysis aimed at quantifying the contribution of each component from both the encoder and decoder sections of the HSCovMamba network. The study focuses on the effects of various configurations of the DLF-Conv, Mamba, convolution layers, and the BDGF-Decoder. It specifically examines the respective impact of the encoder and decoder components on the model's overall performance.

1) Encoder Ablation: Under the condition that the decoder configuration remains consistent with the HSCovMamba model, the encoder component is evaluated through various configurations.

1) *Only Standard Conv (No Mamba, No DLF-Conv):* In this configuration, both the Mamba and DLF-Conv layers are replaced with standard convolution layers in the encoder. This setup serves as a baseline, evaluating the performance of traditional convolution for feature extraction, without the specialized capabilities provided by Mamba or DLF-Conv.
2) *Only Mamba (No DLF-Conv):* Here, only the Mamba layer is retained, with DLF-Conv removed. This configuration evaluates the impact of Mamba's ability to model long-range dependencies and global context, isolating it from the local feature extraction provided by DLF-Conv.
3) *Mamba + Standard Convolution (No DLF-Conv):* This variant replaces the DLF-Conv layer with a standard convolution. It allows for the examination of the effects of using standard convolution for feature extraction, as opposed to the more specialized DLF-Conv layer.
4) *Only DLF-Conv (No Mamba, No Standard Convolution)* In this configuration, only DLF-Conv is used for feature extraction. This test evaluates DLF-Conv's effectiveness in capturing local geometric deformations and texture details, without the contributions of Mamba or standard convolutions.

2) Decoder Ablation: During the evaluation of the decoder's performance, the encoder configuration remains consistent with that of the HSCovMamba model, while multiple variations of the decoder component from HSCovMamba are tested.

1) *Single Path Decoder (No Bi-Directional Fusion):* This configuration uses a single fusion path in the BDGF-Decoder, omitting the bi-directional gated fusion mechanism. It tests the effectiveness of a simplified decoder structure without the full fusion capability.
2) *Bi-Directional Fusion Decoder (No DGFM):* In this version, the BDGF-Decoder is equipped with two fusion paths but without the DGFM modules, where features are directly summed. This evaluates the effect of dual-path fusion without the dynamic modulation of features across the paths.

TABLE V
ABLATION STUDY RESULTS ON PAVIAU, SALINAS, AND HONGHU DATASETS

Model	PaviaU	Salinas	HongHu
HSCovMamba (Full Model)	98.01	97.27	94.96
HSCovMamba Encoder Variations (changes only in encoder)			
Only Standard Conv (No Mamba, No DLF-Conv)	92.30	91.15	89.00
Only Mamba (No DLF-Conv)	95.67	94.10	89.59
Mamba + Standard Conv (No DLF-Conv)	96.45	94.88	92.58
Only DLF-Conv (No Mamba, No Standard Conv)	96.80	95.60	92.71
HSCovMamba Decoder Variations (changes only in decoder)			
Single Path Decoder (No Bi-directional Fusion)	97.00	96.30	93.05
Bi-directional Fusion Decoder (No DGFM)	97.80	96.50	94.53

3) Results: The performance of each ablated version of the HSCovMamba model is summarized in Table V. The results are evaluated based on Kappa (%) on the PaviaU, Salinas, and HongHu HSI datasets. As shown, the full model (with all components) achieves the highest performance, with a Kappa of 98.01% on the PaviaU dataset, 97.27% on the Salinas dataset, and 94.96% on the HongHu dataset, confirming the importance of each component in improving the classification accuracy. Removing the DLF-Conv module leads to a noticeable drop in Kappa, with a decrease to 95.67% on PaviaU, 94.10% on Salinas, and 89.59% in Honghu, indicating that local geometric deformation capture is critical for effective feature extraction. When DLF-Conv is replaced with a standard convolution, the Kappa improves slightly to 96.45% (PaviaU), 94.88% (Salinas), and 92.58% (HongHu), suggesting that although standard convolutions are effective, they do not capture local texture and deformation details as effectively as DLF-Conv. This underscores the importance of the deformable convolution for fine-grained feature extraction. The model using only standard convolution (without Mamba or DLF-Conv) performs the worst, achieving a Kappa of 92.30% on PaviaU, 91.15% on Salinas, and 89.00% on HongHu. This demonstrates the critical role of both Mamba and DLF-Conv in effectively modeling local and global feature dependencies. The removal of the bi-directional fusion paths and the use of a single fusion path slightly reduces performance, with accuracy values of 97.00% (PaviaU), 96.30% (Salinas), and 93.05% (HongHu). When

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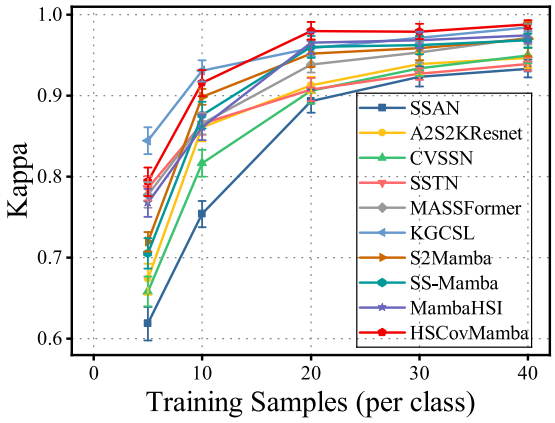


Fig. 7. Effect of training sample numbers on the PaviaU dataset (measured by Kappa).

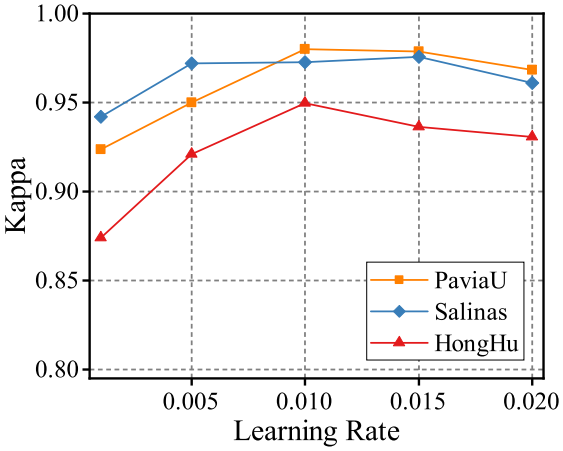


Fig. 8. Effects of the learning rate on classification performance.

the bi-directional fusion paths are retained, but the DGFM module is removed—with a fusion strategy that directly sums the two feature maps—the model performance also decreases slightly compared to HSCovMamba, reaching 97.80% on the PaviaU dataset, 96.50% on the Salinas dataset, and 94.53% on the HongHu dataset. The ablation experiments on the BDGF-Decoder indicate that its mechanism can enhance the feature integration performance.

D. Effect of Training Sample Numbers

Fig. 7 presents the Kappa scores of all methods under varying numbers of training samples for the PaviaU dataset, with standard deviations indicated by error bars. Each result is averaged over five independent runs, with each run using a different training-testing split. As the number of labeled samples increases, all methods show a steady improvement in Kappa scores. Our proposed HSCovMamba model performs exceptionally well when the training data are sufficiently large (e.g., more than 20 samples per class). However, it lags behind KGCSL when the number of labeled samples is scarce (e.g., five or ten per class). This is because KGCSL incorporates geometric and semantic knowledge, which enhances its performance in semi-supervised learning settings. Although it struggles with fewer labeled samples, our method consistently outperforms other alternatives in most settings, showcasing its effectiveness as the training data increases. In summary, HSCovMamba excels in scenarios with a moderate amount of labeled data, where its emphasis on context capture provides a significant advantage.

E. Hyperparameters Analysis

Fig. 8 illustrates the experimental investigation into the impact of different learning rates on the classification performance across three datasets. As the learning rate increases from a very small value, the Kappa scores on all three datasets improve noticeably, indicating that an excessively low learning rate limits effective optimization. The performance reaches its best or near-best level around 0.01, where all datasets achieve relatively strong and stable results. When the learning rate is further increased, a slight performance drop can be observed on all three datasets, suggesting reduced training stability under overly large learning rates. Therefore, 0.01 is selected as the initial learning rate for all three datasets.

F. Complexity and Efficiency Analysis

We conduct a detailed analysis of the computational complexity of HSCovMamba and competing methods on the PaviaU dataset. The evaluation includes floating-point operations (FLOPs), model parameters (Param), inference time, and classification performance measured by Kappa, as summarized in Table VI. As shown in Table VI, the compared methods fall into two inference paradigms, namely patch-based inference and full-scene inference. For patch-based methods, the input size and batch size are kept consistent with the standard settings reported in their original papers since both are architecture-sensitive hyperparameters that directly affect practical inference behavior, which is consistent with recent mainstream research [43], [44] on HSI classification. Although these methods process local patches with relatively large batch sizes, their computational cost still varies considerably across architectures. For example, SSTN and S2Mamba require only 136 M and 594 M FLOPs, respectively, whereas SSAN reaches 239 815 M FLOPs due to its heavy 3-D convolutional operations. By contrast, full-scene methods process the entire HSI in a single pass with batch size 1. Although this inference paradigm generally leads to higher FLOPs than lightweight patch-based models, it often results in substantially faster end-to-end inference. Among the three full-scene methods, HSCovMamba achieves the best overall tradeoff. It obtains the fastest inference time of 0.530 s, outperforming MambaHSI (0.612 s) and KGCSL (2.840 s). Meanwhile, it has the fewest parameters among the full-scene methods, with only 0.152 M parameters, compared with 0.255 M for KGCSL and 0.412 M for MambaHSI. In addition, HSCovMamba achieves the highest Kappa score of 98.01%, surpassing all compared methods. These results demonstrate that HSCovMamba provides a favorable balance between efficiency and accuracy. Although its FLOPs are higher than those of some lightweight patch-based models, its full-scene inference design

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TABLE VI
COMPUTATIONAL EFFICIENCY AND MODEL COMPLEXITY ANALYSIS ON THE PAVIAU DATASET UNDER DIFFERENT INFERENCE PARADIGMS

Metric	Patch-based Methods							Full-scene Methods		
	SSAN	A2S2KResnet	CVSSN	SSTN	S2Mamba	MASSFormer	SS-Mamba	KGCSL	MambaHSI	HSCovMamba
Input size	(7,7)	(9,9)	(9,9)	(9,9)	(11,11)	(13,13)	(27,27)	(610,340)	(610,340)	(610,340)
Batch size	100	32	32	50	64	32	256	1	1	1
FLOPs (M)	239815	5625	1281	**136**	594	3500	18880	17927	13019	12756
Param (M)	24.51	0.221	0.253	**0.016**	0.076	0.640	0.382	0.255	0.412	0.152
Inference time (s)	12.83	19.84	10.96	13.68	15.30	9.940	36.85	2.840	0.612	**0.530**
Kappa (%)	89.33	91.34	90.64	90.79	95.23	93.88	96.03	95.88	96.55	**98.01**

Note: For the patch-based baselines, the input size and batch size are kept consistent with the key hyperparameter settings reported in their original papers.

enables faster prediction in practice, while the lightweight DLF-Conv and efficient decoder help control the model size and computational burden. Therefore, HSCovMamba is well suited for HSI classification scenarios that require both high accuracy and efficient inference.

V. CONCLUSION

This article proposes HSCovMamba, a CNN–Mamba gated-fusion network designed for HSI classification. The architecture consists of two key components. The DLF-Conv is employed for local feature extraction, effectively capturing local textures and fine-grained deformations, and the BDGF-Decoder for efficient multiscale bidirectional fusion. Experiments on three hyperspectral datasets verify that HSCovMamba achieves the highest Kappa accuracy and fastest inference speed while maintaining a compact model size. These merits make it a strong candidate for real-time and high-accuracy HSI classification applications.

Despite the promising results achieved by the proposed method, it still faces limitations, particularly in scenarios with scarce labeled data (e.g., only five samples per class), where its performance is suboptimal. Therefore, future research will investigate semi-supervised learning frameworks to better address the performance bottleneck under extremely limited labeled data.

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